

Capture the Flag

Your Student Portal is now dynamic thanks to interpolation.

But there's a problem.

Your announcement component currently has something like:

```
announcementTitle1 = "Enrollment is Now Open";
announcementBody1 = "Enrollment for the next semester is now
available.";
```

```
announcementTitle2 = "Intramurals 2026";
announcementBody2 = "Registration is now open.";
```

```
announcementTitle3 = "Library Schedule";
announcementBody3 = "The library will be open until 8 PM.";
```

And your HTML has to repeat the same card structure three times.

The school now has **20 announcements**.

Your mission:

Stop repeating yourself. Make Angular generate the cards for you.

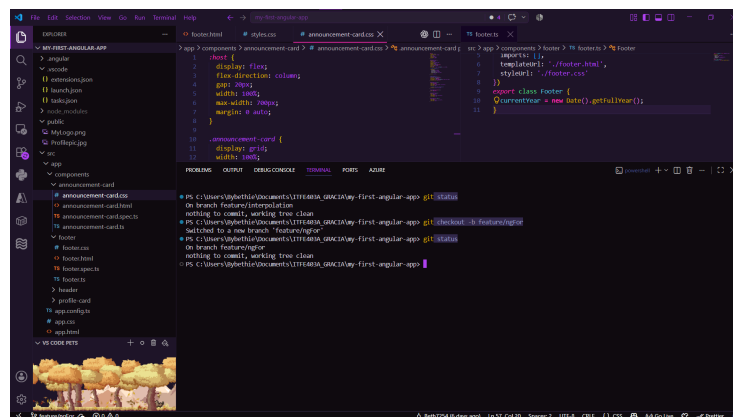
Flag 0 – Mission Branch

Create and switch to a new branch: `feature/ngFor`



Capture

- Show your current Git branch.



FLAG 1 — The Repetition Problem

Open your announcement component.

Look at your existing announcement cards.

You should currently have something similar to:

```
<div class="announcement-card">
  <h3>{{ announcementTitle1 }}</h3>
  <p>{{ announcementBody1 }}</p>
</div>
```

```
<div class="announcement-card">
  <h3>{{ announcementTitle2 }}</h3>
  <p>{{ announcementBody2 }}</p>
</div>
```

```
<div class="announcement-card">
  <h3>{{ announcementTitle3 }}</h3>
  <p>{{ announcementBody3 }}</p>
</div>
```

Mission

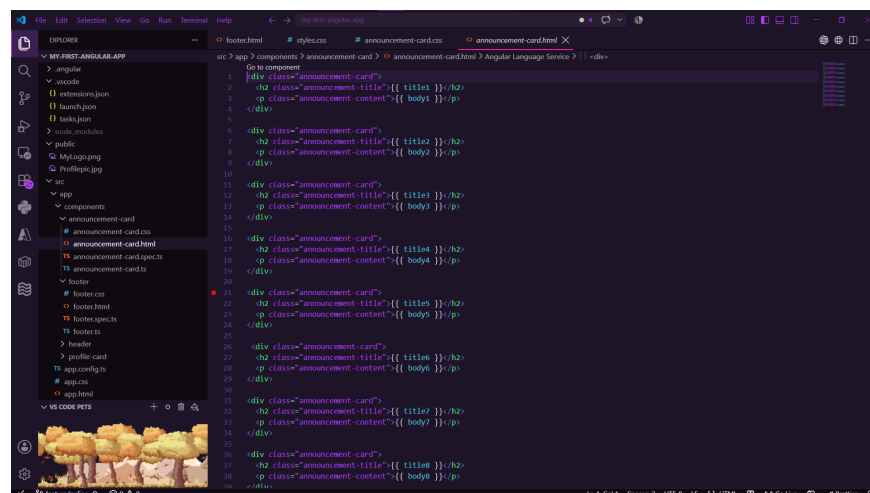
Imagine the school gives you **10 more announcements**.

Would you want to copy and paste the card **10 more times**?

Don't change anything yet.



Capture: Your repeated announcement cards.



FLAG 2 — Put the Data Together

Instead of creating separate variables, create **one array**.

Research Hint

An array can contain multiple objects.

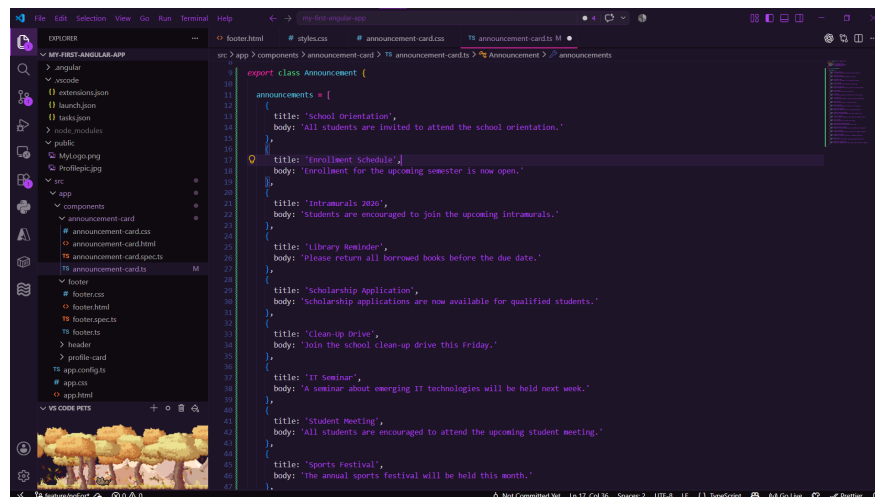
You don't need to memorize the syntax.

Search:

Angular array of objects



Capture: Your **announcements** array.



FLAG 3 — Meet *ngFor

Now comes the important part.

You have:

```
announcements = [  
  { ... },  
  { ... },  
  { ... }  
];
```

How can Angular create one card **for every item** in that array?

Hint

Research:

**Angular ngFor*

You should discover syntax similar to:

**ngFor="let announcement of announcements"*

Your mission is to use it on your announcement card.

Instead of having:

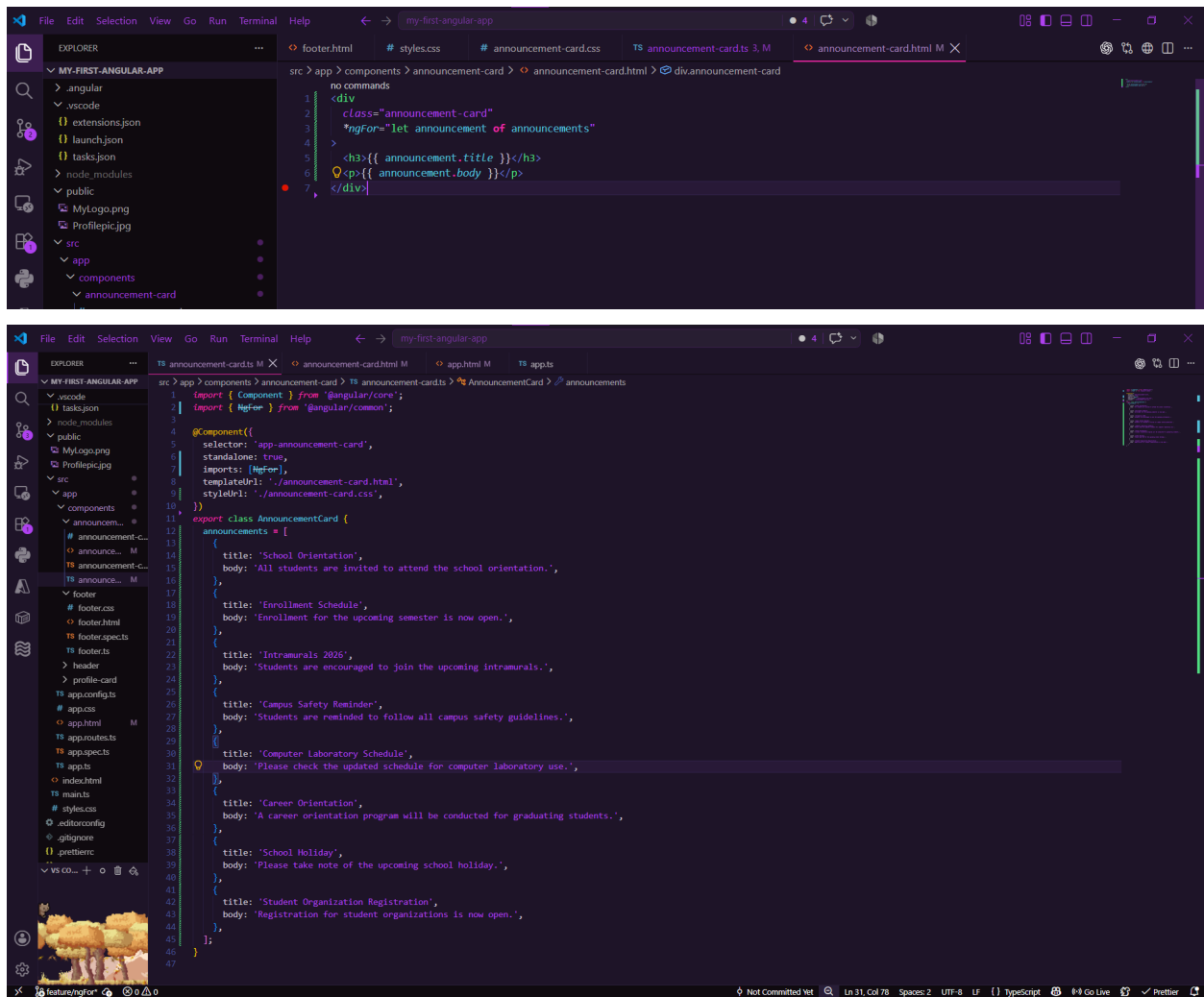
<div class="announcement-card">

repeated three times, you should have **one card template** that Angular repeats.



Capture:

- Your updated code.



FLAG 4 — Display the Current Announcement

You now have a loop variable.

Inside the repeated card, display title and body:

Your browser should now display all eight announcements.

Important

You should have:

1 array

1 card template

8 displayed cards

Not:

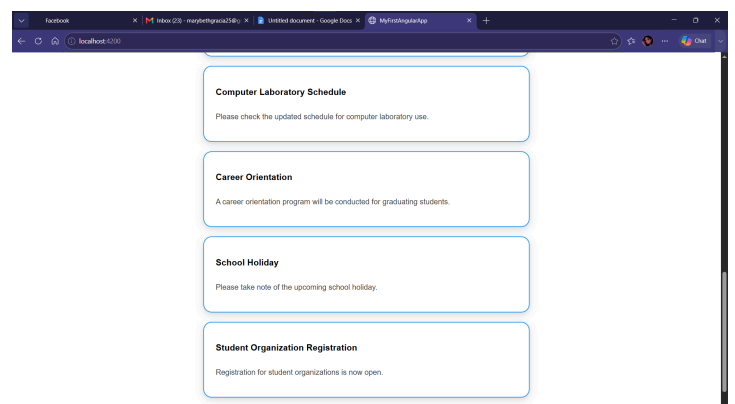
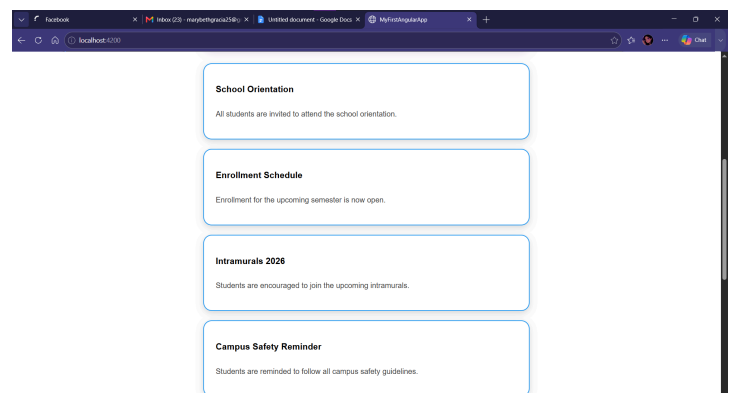
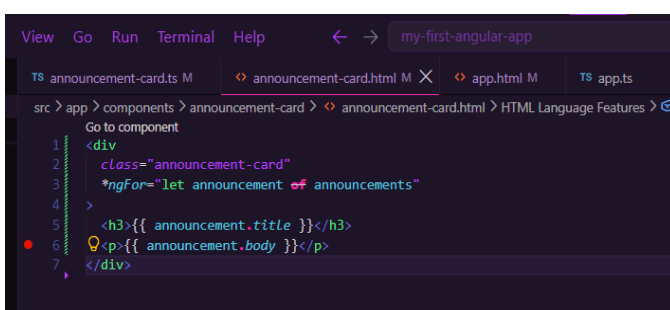
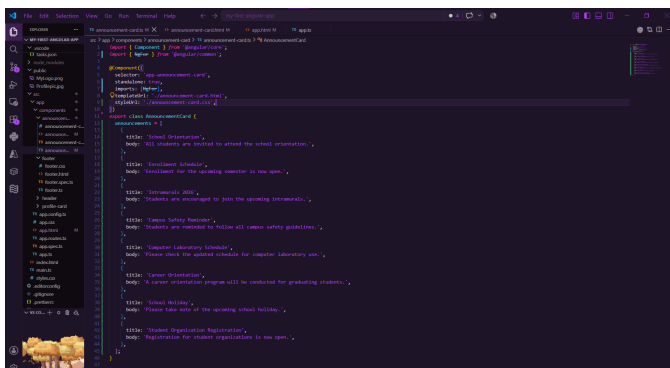
8 arrays

8 card templates



Capture:

- Browser showing all announcements.
- Updated code.



FLAG 5 — Add More Data

Now test whether your loop actually works.

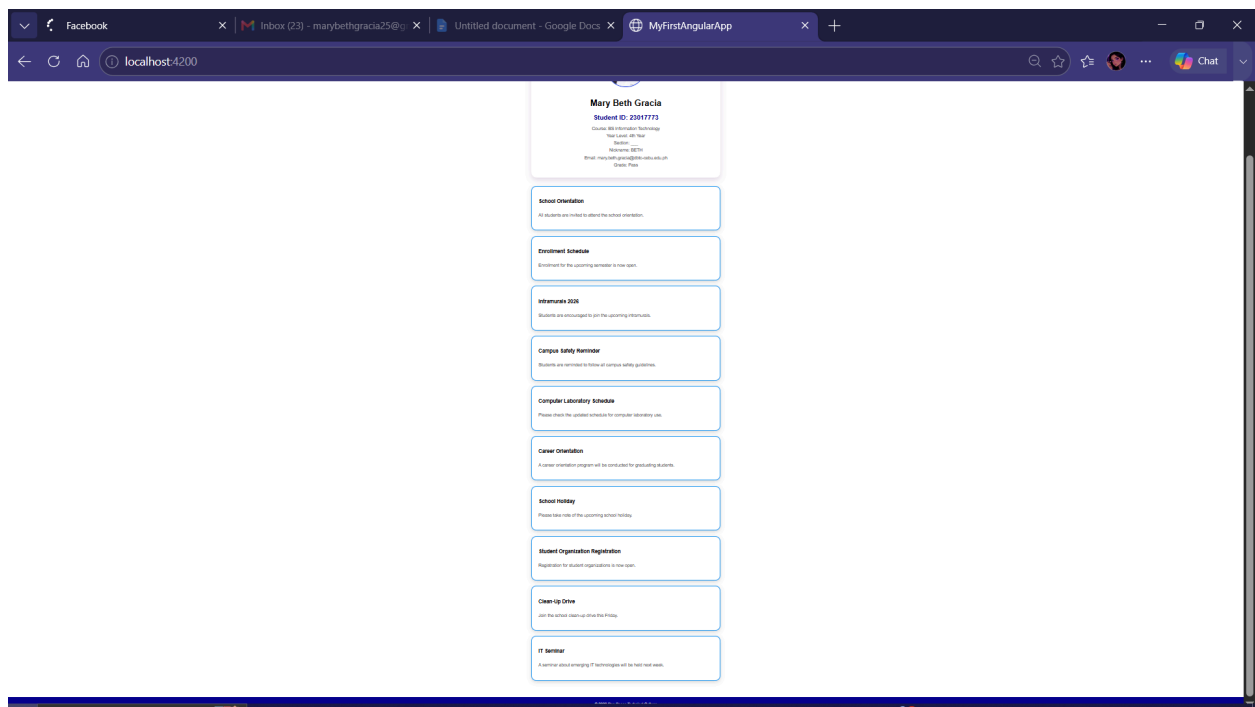
Add **two more announcements** to the array.

You should **not change your HTML**.

If two additional cards automatically appear:

🚩 **FLAG CAPTURED**

📸 **Capture:** Browser showing all ten announcements.



FLAG 6 — Make the Cards More Realistic

Your announcement objects currently contain only:

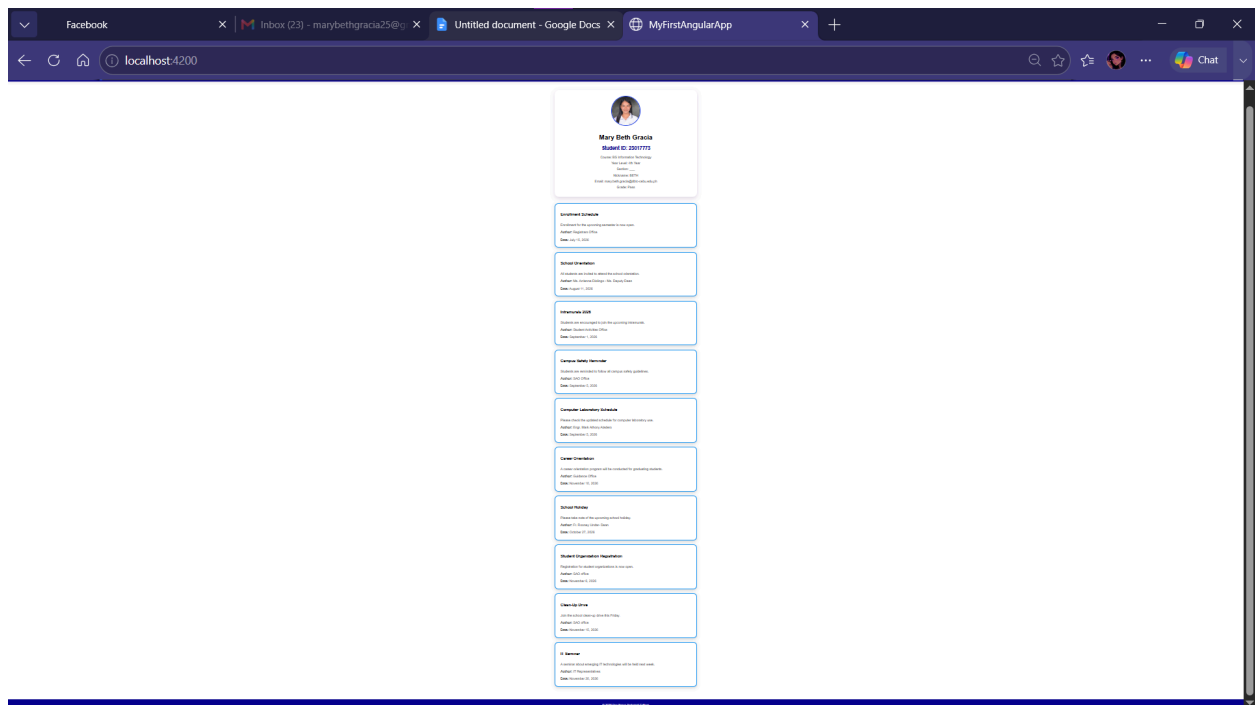
title
body

Add more information: author and date.

Update your card to display all fields.

Use **interpolation** for all values.

 **Capture:** Your richer announcement cards.



FLAG 7 — Student List

Announcements aren't the only thing that can be repeated.

Create another array named students.

Create a simple student card.

Use ***ngFor** to display every student.

Expected example:

Juan Dela Cruz
BSIT • 2nd Year

Maria Santos
BSIT • 3rd Year

Pedro Garcia
BSCS • 1st Year

 **Capture:** Student list.

Author: IT Representatives Date: November 20, 2026

Student List

Juan Dela Cruz
BSIT • 2nd Year

Maria Santos
BSIT • 3rd Year

Pedro Garcia
BSCS • 1st Year

FLAG 8 — One Template, Many Students

Add another student to your array.

Do **not** modify the HTML.

The new student should automatically appear.

This is the important concept:

DATA CHANGES



ARRAY CHANGES



ANGULAR GENERATES THE UI

 **Capture:** Before and after adding the student.

Student List

Juan Dela Cruz
BSIT • 2nd Year

Maria Santos
BSIT • 3rd Year

Pedro Garcia
BSCS • 1st Year

Student List

Juan Dela Cruz
BSIT • 2nd Year

Maria Santos
BSIT • 3rd Year

Pedro Garcia
BSCS • 1st Year

Syvy Sarvida
BSIT • 4th Year

FLAG 9 — Research Challenge

I will **not** provide the next syntax.

Research:

**Angular ngFor index*

Find out how to access the position of the current item.

Use it to display:

#1 Juan Dela Cruz

#2 Maria Santos

#3 Pedro Garcia

Without hardcoding the numbers.

Hint:

let i = index



Capture:

- Your numbered student list.
- Your code

```
src > app > components > announcement-card > announcement-card.html > ...
Go to component
1 <div class="announcement-card" *ngFor="let announcement of announcements">
2   <h2>{{ announcement.title }}</h2>
3
4   <p>{{ announcement.body }}</p>
5
6   <span> <strong>Author:</strong> {{ announcement.author }} </span>
7
8   <span> <strong>Date:</strong> {{ announcement.date }} </span>
9 </div>
10
11
12 <div class="student-list">
13   <h2>Student List</h2>
14
15   <div
16     class="student-card"
17     *ngFor="let student of students; let i = index">
18     <h3>#{{ i + 1 }} {{ student.name }}</h3>
19     <p>{{ student.course }} • {{ student.year }}</p>
20   </div>
21 </div>
22
```

```
students = [
  {
    name: 'Niel Jhon Celocia',
    course: 'BSIT',
    year: '4th Year',
  },
  {
    name: 'Jownie Ann Malate',
    course: 'BSIT',
    year: '4th Year',
  },
  {
    name: 'Bonna Mae Pitogo',
    course: 'BSIT',
    year: '4th Year',
  },
  {
    name: 'Kurt Wojtyle Rizal',
    course: 'BSIT',
    year: '4th Year',
  }
]
```

Student List

#1 Niel Jhon Celocia

BSIT • 4th Year

#2 Jownie Ann Malate

BSIT • 4th Year

#3 Bonna Mae Pitogo

BSIT • 4th Year

#4 Kurt Wojtyle Rizal

BSIT • 4th Year

FLAG 10 — Empty List

Now test what happens when there are no announcements.

Change:

```
announcements = [];
```

What happens?

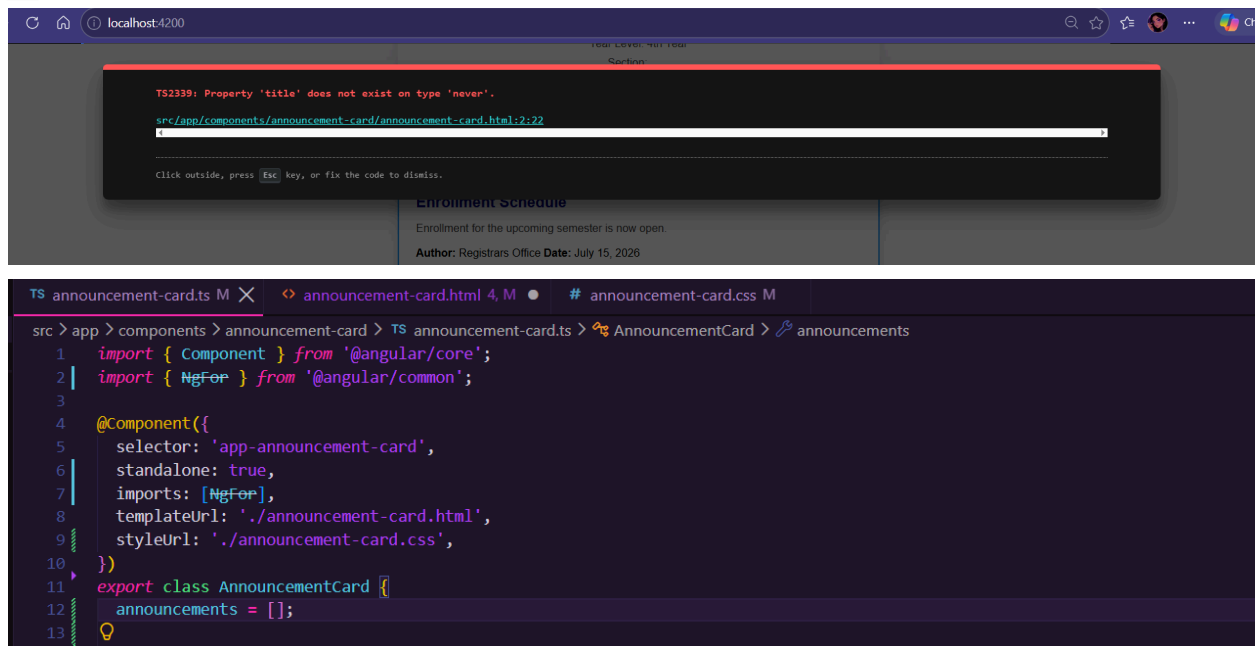
Write a short observation in your documentation:

What happens when the array is empty?

Answer: *It displays an error, and can't properly run the code.*

Then restore your announcements.

 **Capture:** Your experiment and observation.



FLAG 11 — Dynamic Student Portal

Your Student Portal should now contain:

 **Your Student Profile**

Still using interpolation:

Name

Student ID
Course
Year
Grade
Academic Standing
Email

Announcements

Generated from:

announcements = [...]

Using:

*ngFor

Student List

Generated from:

students = [...]

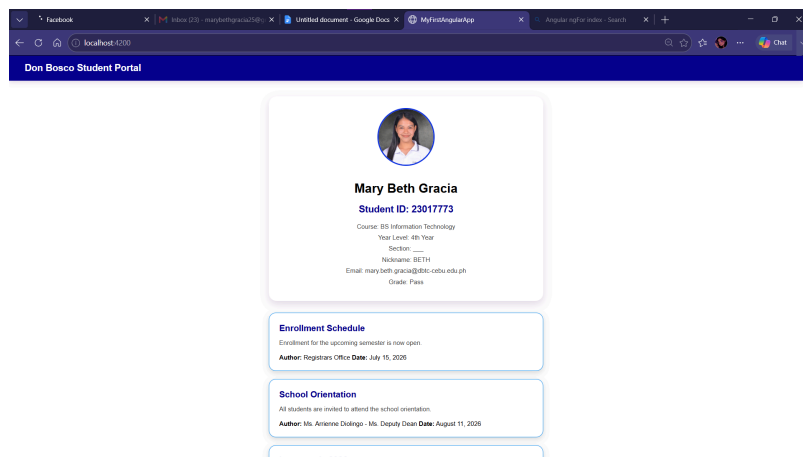
Using:

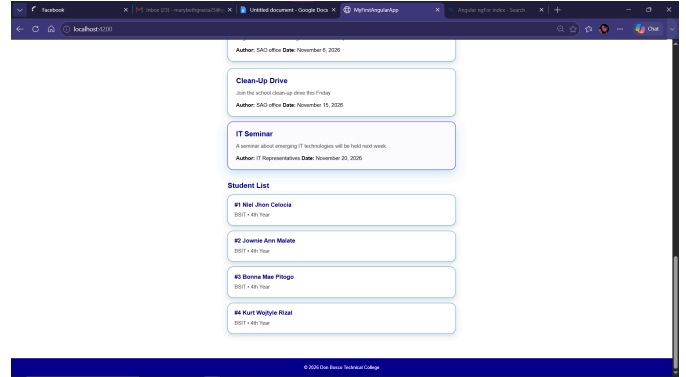
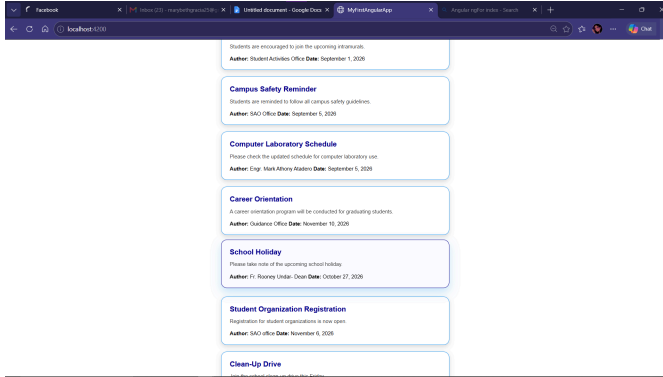
*ngFor and index

Your final requirement

Final Screenshot

- Entire browser UI of all requirements in this activity





FLAG 12 — Commit & Push

Final Screenshot

- Successful push to feature/ngFor